

WEEK-3 SKILL

Task 1: To Create Main Loop, Display Prompt, Read User Input, Handle Exit Conditions, Design Control Flow Diagram, Test Interactive Loop

```
root@Rahul:~# nano week3.c
root@Rahul:~# |
```

```
#include <stdio.h>
#include <string.h>

int main()
{
    char command[100];

    while (1)
    {
        printf("myshell> ");

        fgets(command, sizeof(command), stdin);

        command[strcspn(command, "\n")] = '\0';

        if (strcmp(command, "exit") == 0)
        {
            printf("Exiting shell...\n");
            break;
        }

        printf("You entered: %s\n", command);
    }

    return 0;
}
```

OUTPUT

```
root@Rahul:~# nano week3.c
root@Rahul:~# gcc week3.c -o week3
root@Rahul:~# ./week3
myshell> Hello
You entered: Hello
myshell> welcom
You entered: welcom
myshell> exit
Exiting shell...
root@Rahul:~# |
```

Design Control Flow Diagram

```
Start
|
Display Prompt
|
Read User Input
|
Is Input "exit"?
|   |
No   Yes
|   |
Display Exit Program
Input   |
|       |
|       |
<-----
```

Task 2: Using the following system calls, copy the content from File1.txt to File2.txt.

Step 1: Create File1

```
nano File1.txt
```

Write:

Welcome to Ubuntu Lab

Step 2: Create File2

```
root@Rahul:~# nano file.txt
root@Rahul:~# touch File2.txt
root@Rahul:~# |
```

Step 3: Create C Program

```
root@Rahul:~# nano file.txt
root@Rahul:~# touch File2.txt
root@Rahul:~# nano copyfile.c|
```

```

#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
int main()
{
    char buffer[100];
    int source, destination, bytes;
    // open() - Opens File1 for reading
    source = open("File1.txt", O_RDONLY);
    // open() - Opens File2 for writing
    destination = open("File2.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);
    // read() and write() are used inside the loop
    while((bytes = read(source, buffer, sizeof(buffer))) > 0)
    {
        write(destination, buffer, bytes);
    }
    // close() - Closes both files
    close(source);
    close(destination);
    printf("File copied successfully.\n");
    return 0;
}

```

Compile and Run

```

root@Rahul:~# nano file.txt
root@Rahul:~# touch File2.txt
root@Rahul:~# nano copyfile.c
root@Rahul:~# gcc copyfile.c -o copyfile
root@Rahul:~# ./copyfile

```

OUTPUT

```

root@Rahul:~# nano file.txt
root@Rahul:~# touch File2.txt
root@Rahul:~# nano copyfile.c
root@Rahul:~# gcc copyfile.c -o copyfile
root@Rahul:~# ./copyfile
File copied successfully.
root@Rahul:~#

```

